

tmux documentation

tmx-state

Revision: 1 **Last modified:** 2026-05-22T12:20:00Z **Authority:** vasic-digital tmux project
Maintainer: milosvasic **Scope:** §11.4.18 script companion doc for the `tmx-state` Go binary
(source: `scripts/tmx-state/`, built artefact: `scripts/tmx-state-bin`)

Purpose

Per-session cwd persistence daemon for the `tmx` wrapper. Records the last-known working directory each `tmux` session was in when its client detached or the session closed, so the next `tmx new -s NAME` can spawn the session back in that directory. Atomic state writes (`temp + rename`

- `fsync`) under an `fcntl F_SETLKW` advisory lock survive N concurrent panes recording simultaneously without data loss. Design authority: `docs/superpowers/specs/2026-05-22-tmx-shell-session-resume-design.md §4.B + §5.1 + §5.3`.

Usage

```
scripts/tmx-state-bin record <session> <abs-path>
scripts/tmx-state-bin recall <session>
scripts/tmx-state-bin list
scripts/tmx-state-bin forget <session>
scripts/tmx-state-bin version
```

Concrete examples:

```
# Record that session 'demo' was last in /tmp.
scripts/tmx-state-bin record demo /tmp
```

```
# Recall it (prints "/tmp" to stdout, no trailing newline).
scripts/tmx-state-bin recall demo
```

```
# Use inside a subshell expansion (the canonical wrapper
  pattern):
LAST_PWD="$(scripts/tmx-state-bin recall demo 2>/dev/null ||
  true)"
START_DIR="${LAST_PWD:-$HOME}"
```

```
# Tab-separated listing.
scripts/tmx-state-bin list
# → demo<TAB>/tmp<TAB>1748000000
```

```
# Drop a session's record (idempotent — exit 0 whether present or
  not).
```

```
scripts/tmx-state-bin forget demo
```

```
# Version probe.
```

```
scripts/tmx-state-bin version      # → tmx-state v1.0.9
```

Inputs

- **Positional args** as above.
- `$TMX_STATE_FILE` (optional) — overrides the default state file path. Default: `$HOME/.tmx/state.json`. Used by tests to point at a temp file rather than the operator's real state.
- `$HOME` — used to derive the default state file path.

Outputs

- **stdout** — `recall` prints `last_pwd` (no trailing newline) on hit; `list` prints tab-separated rows; `version` / `help` print their banner. Other subcommands are silent on success.
- **stderr** — error messages on failure; one-line notice when a corrupt state file is auto-rebuilt during `record`.
- **state file** at `$TMX_STATE_FILE` (or `~/.tmx/state.json`) — JSON document, schema below. Parent directory is created with mode 0700; file is created with mode 0600.

State file JSON schema (from spec §5.3, verbatim)

```
{
  "schema_version": 1,
  "sessions": {
    "work": {
      "last_pwd": "/Users/milosvasic/Projects/tmux",
      "last_seen_unix": 1748000000,
      "created_unix": 1747000000,
      "host": "nezha.local"
    }
  }
}
```

Single-file design (not jsonl) — fits in memory trivially (kilobytes for hundreds of sessions), allows atomic write+rename, and is easy to inspect by hand. `created_unix` is preserved across `record` upserts; `last_seen_unix` and `last_pwd` are overwritten each call.

Exit codes

Code	Subcommand	Meaning
0	any	success
1	record / list / forget	usage error, IO error, permission denied
1	recall	session not present in state

Code	Subcommand	Meaning
2	<code>recall</code>	state file unreadable (corrupt / IO error) — distinguishes "no record" from "cannot read" so callers can fall back deliberately rather than treat both as "no record"

Side-effects

- Creates `~/ .tmx/` mode 0700 if missing.
- Creates/overwrites `~/ .tmx/state.json` mode 0600 atomically (`temp + fsync + rename`).
- Holds an advisory `fcntl F_SETLK` lock on the state file for the duration of every mutating call. Concurrent `record` invocations serialise; latest-wins per spec §6 edge case 8.
- On corrupt state file: rebuilds to `{schema_version: 1, sessions: {}}`, emits a one-line `stderr` notice during `record`; for `recall` returns exit 2 so callers can choose to fall back rather than silently treat the empty rebuild as "no record".

Dependencies

- Go $\geq$ 1.21 toolchain (build-time only; produced binary is static).
- POSIX filesystem semantics (`rename(2)` atomicity, `fcntl(2)` advisory locking).
- No network. No external services. No third-party Go modules.

Cross-references

- Design spec: `docs/superpowers/specs/2026-05-22-tmx-shell-session-resume-design.md`
- Operator user guide: `docs/guides/tmx-state.md` (P8 deliverable)
- Wrapper integration: `scripts/tmx.template` (P4 — calls `recall` on `tmx new` and installs `record`-firing `tmux` hooks on `detach/close`)
- Shell init: `scripts/tmx-shell-init.sh` (P2 — operator-facing prompt that ultimately calls the wrapper)
- SSH dispatch: `scripts/tmx-ssh-dispatch.sh` (P3 — `ssh host-tmx X path` also queries `recall`)
- Constitution: §11.4.18 (script docs), §11.4.5 (captured evidence), §11.4.50 (deterministic consistency — concurrent `record` test), §11.4.65 (universal Markdown export — this doc gets `.html + .pdf` siblings via `scripts/testing/sync_all_markdown_exports.sh`)

Last verified

2026-05-22 — anti-bluff dry-trace of the wrapper integration captured in `scripts/tmx.template` PWU P4 evidence; full on-device run gated on PWU P6 test `18_state_persistence.sh`.